

PUMPING LEMMA FOR CF LANGUAGES

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AP/CSE

LEARNING OBJECTIVE

- To Design pushdown automata for any CFL (K3)
 - To Understand the concept of pumping lemma for CFL

STATEMENT OF THE CFL PUMPING LEMMA

For every context-free language L

There is an integer n, such that

For every string w in L of length $\geq n$

There exists $w = uvxyz$ such that:

1. $|vxy| \leq n$.
2. $|vy| > 0$.
3. For all $i \geq 0$, $uv^i xy^i z$ is in L.

$\underbrace{\text{CFL}}_{\text{PDA} \rightarrow \text{CFG}}$
 n
 $|w| \geq n$
 $w = uvxyz$
 $|vxy| \leq n$
 $vy \neq \epsilon \quad |vy| > 0$
 $\forall i \quad uv^i xy^i z \in L$

n
 $|w| \geq n$
 $w = xyz$
 $|xy| < n$
 $xy \neq \epsilon$
 $|y| > 0$
 $xy^i z \in L \rightarrow \text{PL}$
 $\left. \begin{array}{l} \text{DFA} \\ \text{NFA} \\ \text{E-NFA} \\ \text{RE} \end{array} \right\} \text{RL}$

EXAMPLE

- $L = \{a^n b^n c^n \mid n \geq 0\}$

$$\begin{aligned} |w| &\geq n \\ w &= uvxyz \\ |vxy| &\leq n \end{aligned}$$

$$|vy| > 0 \quad \& \quad vy \neq \epsilon$$

$$uv^i xy^i z \in L$$

$$w = a^p b^p c^p$$

$$\begin{aligned} |w| &\geq n \\ w &= a^p \\ vxy &= b^p \\ vy &= b^{p-r} \\ z &= c^p \end{aligned}$$

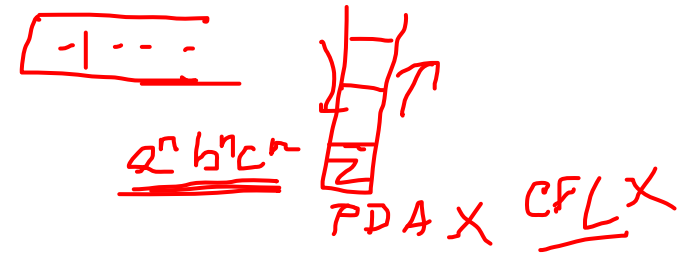
$$\begin{aligned} |vxy| &< n \\ |vy| &> 0 \end{aligned}$$

$$\begin{aligned} uv^i xy^i z &= u v x y (vy)^{i-1} z \\ &= a^p b^p (b^{p-r})^{i-1} c^p \end{aligned}$$

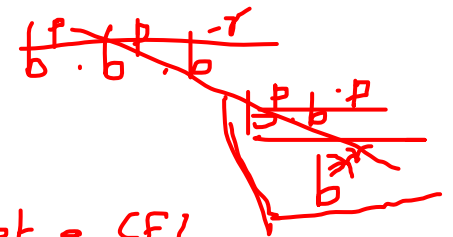
$$\begin{aligned} i=0 & a^p b^p (b^{p-r})^{-1} c^p \\ &= a^p \cancel{b^p} \cancel{b^p} b^r c^p \\ &= a^p b^r c^p \notin L \end{aligned}$$

$$\begin{aligned} i=1 & a^p b^p (b^{p-r})^0 c^p \\ &= a^p b^p c^p \in L \end{aligned}$$

$$\begin{aligned} i=2 & a^p b^p (b^{p-r})^1 c^p \\ &= a^p b^p \cdot b^p \cdot b^{-r} c^p \\ &= a^p b^{2p-r} c^p \notin L \end{aligned}$$



$$\begin{aligned} u & \\ vxy & \\ vy & z \end{aligned}$$



$\therefore L$ is not a CFL

EXAMPLE

- $L = \{a^k b^j c^k d^j \mid k, j \geq 1\}$

$$\begin{aligned}
 w &= a^p \\
 vxy &= b^q \\
 |vxy| &< n \\
 vy &= b^{q-h} \\
 z &= c^p d^q \\
 \underline{uv^i xy^i z} \\
 &= u vxy (vy)^{i-1} z \\
 &= a^p b^q (b^{q-h})^{i-1} c^p d^q
 \end{aligned}$$

$$\begin{aligned}
 \underline{i=0} \\
 uv^0 xy^0 z &= a^p b^q (b^{q-h})^{-1} c^p d^q \\
 &= a^p b^q b^{-q} b^h c^p d^q \\
 &= a^p b^h c^p d^q \notin L \\
 \\
 \underline{i=1} \\
 uv^1 xy^1 z &= a^p b^q (b^{q-h})^0 c^p d^q \\
 &= a^p b^q c^p d^q \in L \\
 \\
 \underline{i=2} \\
 uv^2 xy^2 z &= a^p b^q (b^{q-h})^{2-1} c^p d^q \\
 &= a^p b^q b^{q-h} b^{q-h} c^p d^q \\
 &= a^p b^{2q-h} c^p d^q \notin L
 \end{aligned}$$

$\therefore$ The language is not a CFL.

EXAMPLE

- $L = \{a^n b^{n+1} c^{n+2} \mid n \geq 0\}$

INTUITION

- Recall the pumping lemma for regular languages.
- It told us that if there was a string long enough to cause a cycle in the DFA for the language, then we could “pump” the cycle and discover an infinite sequence of strings that had to be in the language.

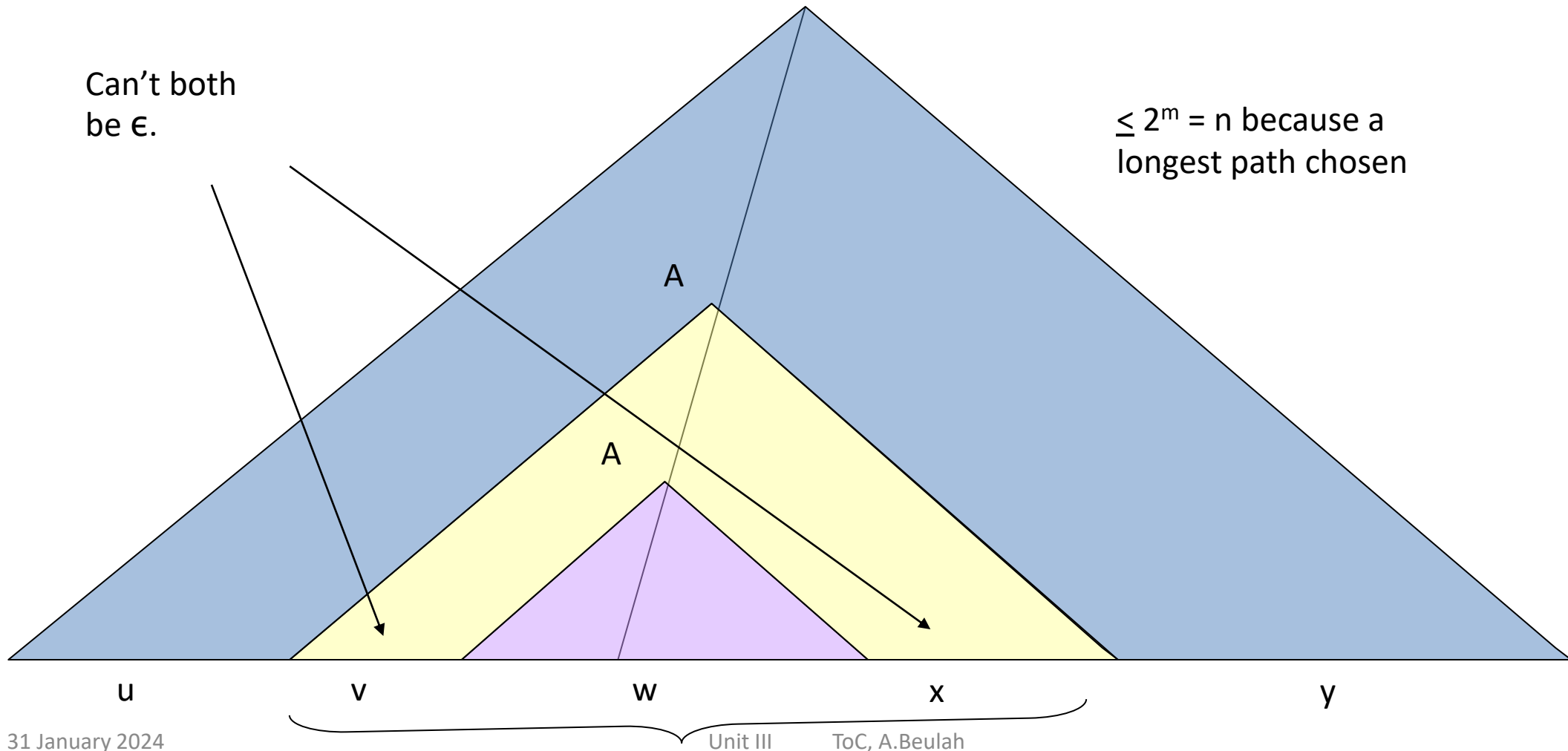
INTUITION – (2)

- For CFL's the situation is a little more complicated.
- We can always find **two** pieces of any sufficiently long string to “pump” in tandem.
 - **That is**: if we repeat each of the two pieces the same number of times, we get another string in the language.

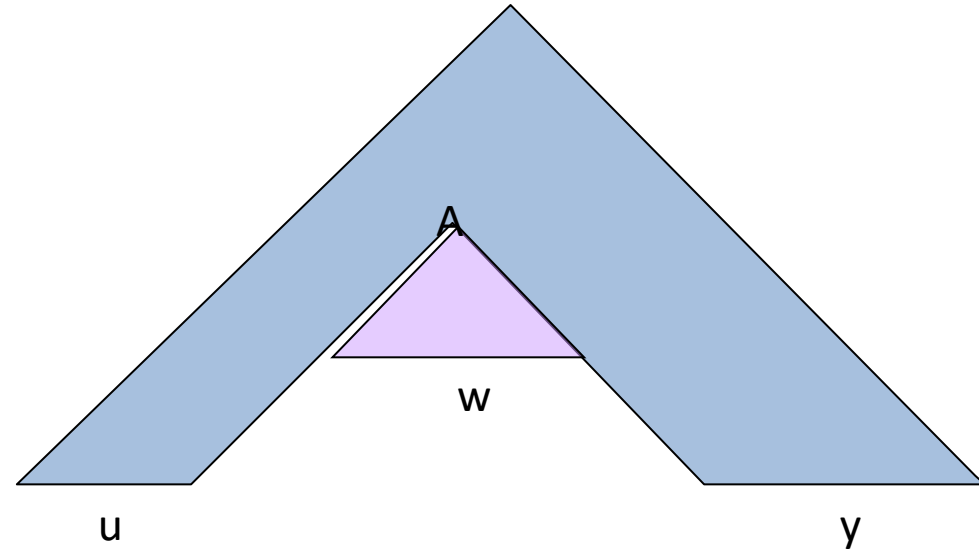
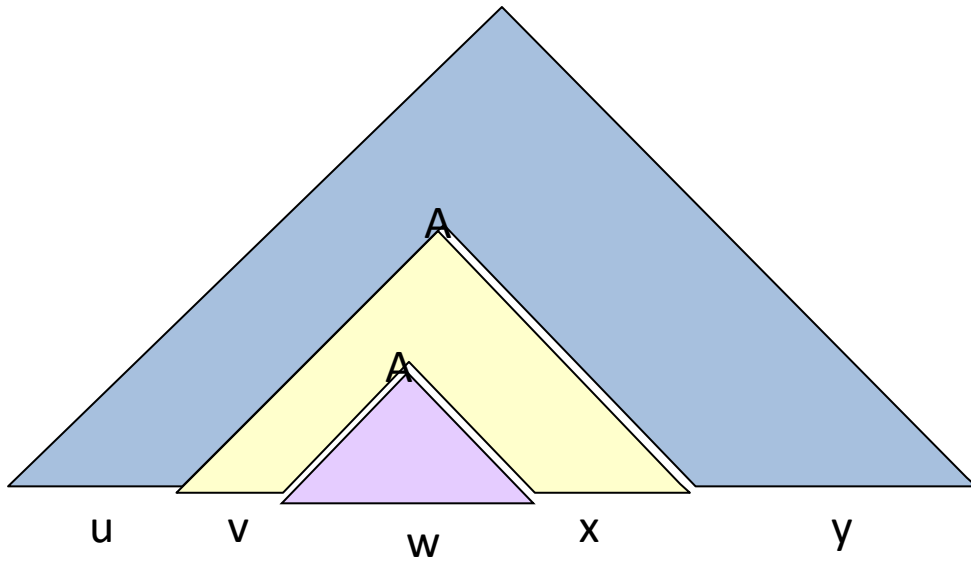
PROOF OF THE PUMPING LEMMA

- Let the grammar have m variables.
- Pick $n = 2^m$.
- Let $|z| \geq n$.

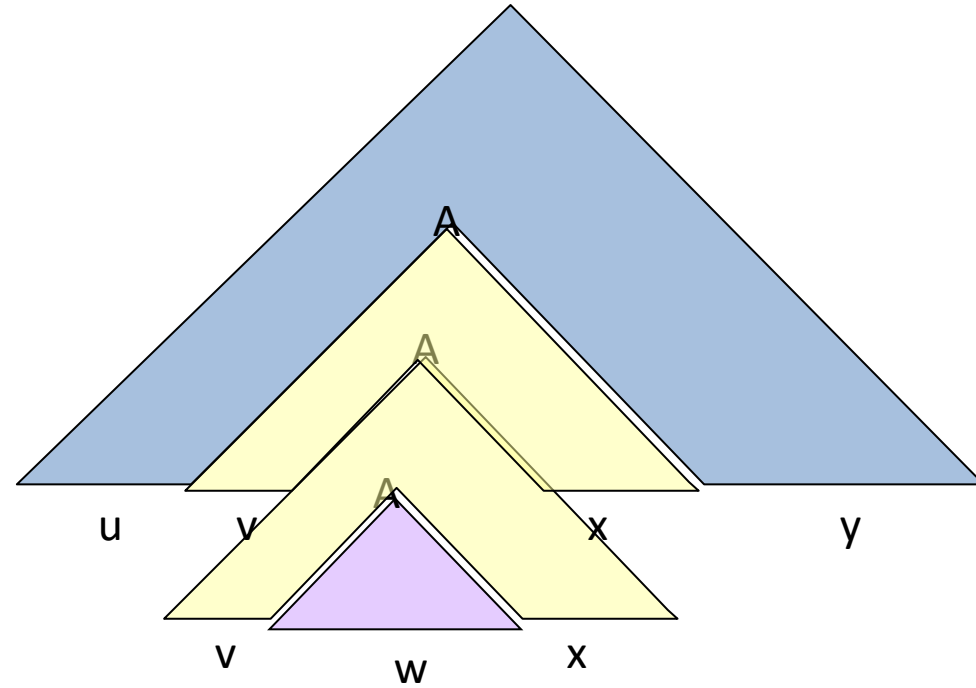
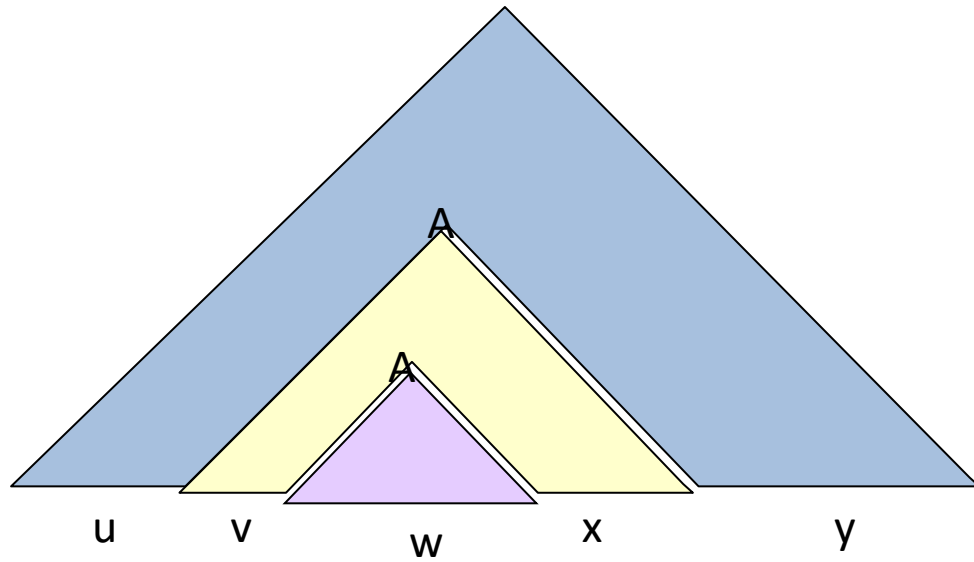
Parse Tree in the Pumping-Lemma **Proof**



Pump Zero Times

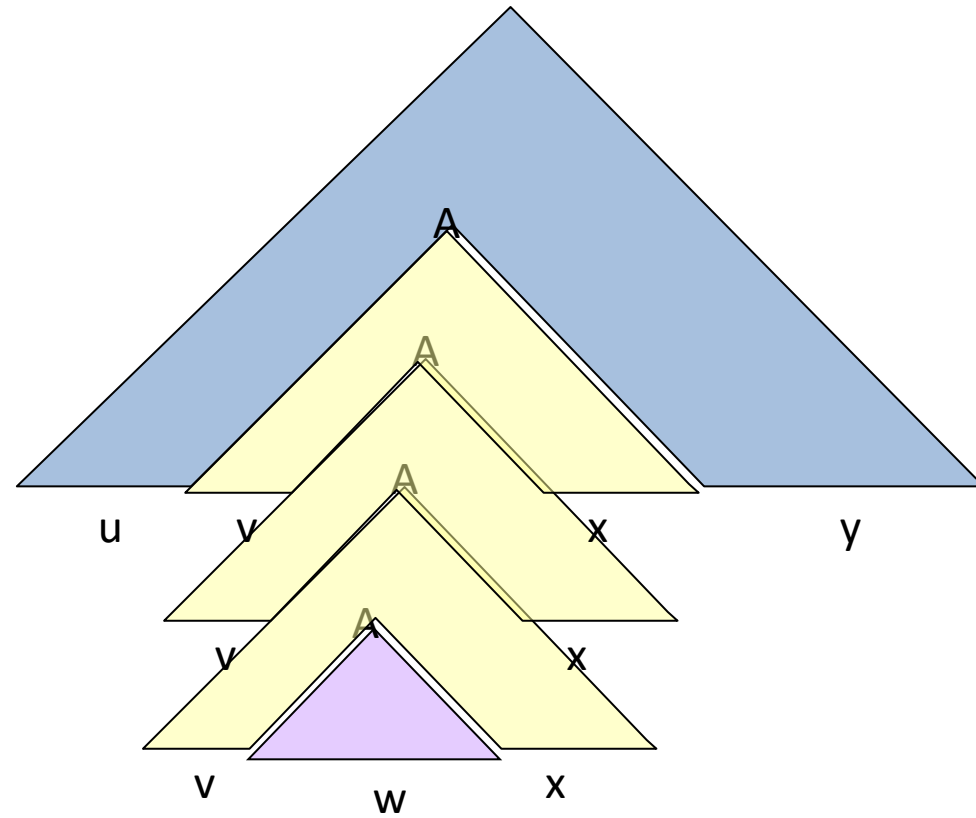
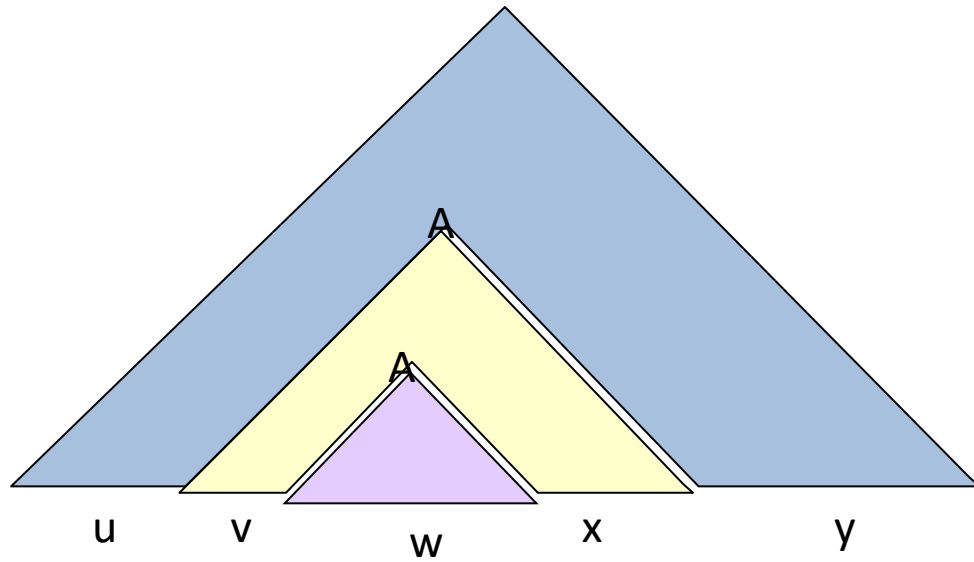


Pump Twice



Pump Thrice

Etc., Etc.



USING THE PUMPING LEMMA

- Non-CFL's typically involve trying to match two pairs of counts or match two strings.
- **Example:** The text uses the pumping lemma to show that $\{ww \mid w \text{ in } (0+1)^*\}$ is not a CFL.

- $\{0^i 1 0^i \mid i \geq 1\}$ is a CFL.
 - We can match one pair of counts.
- But $L = \{0^i 1 0^i 1 0^i \mid i \geq 1\}$ is not.
 - We can't match two pairs, or three counts as a group.
- **Proof** using the pumping lemma.
- Suppose L were a CFL.
- Let n be L 's pumping-lemma constant.

- Consider $z = 0^n 1 0^n 1 0^n$.
- We can write $z = uvwxy$, where $|vwx| \leq n$, and $|vx| \geq 1$.
- **Case 1:** vx has no 0's.
 - Then at least one of them is a 1, and uw has at most one 1, which no string in L does.

- Still considering $z = 0^n 1 0^n 1 0^n$.
- **Case 2:** vx has at least one 0.
 - vwx is too short ($\text{length} \leq n$) to extend to all three blocks of 0's in $0^n 1 0^n 1 0^n$.
 - Thus, uw has at least one block of n 0's, and at least one block with fewer than n 0's.
 - Thus, uw is not in L .

SUMMARY

- Pumping lemma for Context free languages
 - Lemma – to prove the given language is not CFL
 - Problems
 - Proof

TEST YOUR KNOWLEDGE

- Pumping lemma for context free grammar is used for
 - a) Proving certain languages are not context free
 - b) Proving language is infinite
 - c) Both (a) and (b)
 - d) None of these

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008